

Submodule pitfalls

A submodule embeds another repository at a pinned commit, the parent stores a gitlink, not a copy of every file

Init, detached HEAD, pin bumps

- After a normal clone
- Inside the submodule, detached HEAD is normal
- To move the pin
- Push the submodule commit to its remote first if needed, then push the parent
- Two repos, two push steps, easy to forget one

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Submodule pitfalls

Practice `init`/`update`, detached HEAD inside `vendor/dv-lib`, and committing the parent pin after a bump.

Starter example: You cloned the chip repo — `vendor/dv-lib` is an empty submodule folder until submodule `update --init`. Then watch detached HEAD and pin-update pitfalls.

Load starter example

Challenges 0 / 22 cleared

Quiz: empty: After plain clone, submodule dirs are often? Answer: empty

Answer

Show hintCheckNextIdle

Quiz: emptyQuiz: initStarter emptyInit updateQuiz: detachedSee detachedSwitch branchQuiz: pinBump dirtyCommit pinQuiz: push orderForget push trapFix forgetQuiz: cloneQuiz: why detachFull bump flowCommit before pushQuiz: gitmodulesMatch after initQuiz: CI Bump re-detachesStarter pin

Classic pitfalls

Forgotten init

Clone leaves submodule empty until `update --init`.

Detached HEAD

Submodule checks out a pinned SHA, not a branch.

Pin updates

Bump inside submodule, then commit the parent gitlink.

Fresh clone (empty)InitializedPin bump pending

Submodule pitfalls lab starter

Real Git practice

```
wsl - module18-submodule-pitfalls/examples/submodules

# Track A - real Unix
# real shell session (Track A)

# practice sandbox: /tmp/git-sm-NyXgPh

$ git clone <parent> fresh-clone

$ ls vendor/dv-lib

$ git submodule status
-b8f039e683b82cd10213250df81ea9c53fadfecc vendor/dv-lib

$ git submodule update --init --recursive

$ cat vendor/dv-lib/VERSION
v1

$ git status
## HEAD (no branch)

$ git fetch

$ git checkout 5ebdb87ff733414fd2ec99f562c49d06f5adfc

$ git add vendor/dv-lib

$ git commit -m "Bump dv-lib pin"

$ git submodule status
5ebdb87ff733414fd2ec99f562c49d06f5adfc vendor/dv-lib (heads/main)
```

Real shell — init, detached HEAD, bump pin

Real Git practice — try these

```
# (setup) parent repo pins vendor/dv-lib at lib v1
```

```
# git clone <parent> fresh-clone — clone without submodules populated  
git clone file:///.../parent fresh-clone
```

```
# ls vendor/dv-lib — empty until init  
ls vendor/dv-lib
```

```
# git submodule status — minus prefix means not initialized  
git submodule status
```

```
# git submodule update --init --recursive — populate submodule  
checkout  
git submodule update --init --recursive
```

```
# cat vendor/dv-lib/VERSION — library files appear
```

Pitfalls to watch

- An empty vendor or external folder after clone almost always means forgotten init
- Do not commit only inside the submodule and stop; the parent must record the new SHA
- Detached HEAD inside a submodule is expected until you deliberately switch a branch for
- CI should clone with recurse submodules or run init in the build script
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and fix init, detached HEAD, and pin-update scenarios
- On real Git, clone, init, inspect status inside the submodule, and bump the parent pin
- When you are ready